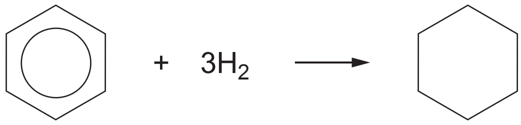
Examiner only

SECTION B

Answer **all** questions.

8. (a) The starting material for making polyamides is often benzene, obtained from petroleum.

(i) The first stage in the production of a particular polyamide is the hydrogenation of benzene at increased temperature and pressure.



Suggest the name of a catalyst that can be used for the hydrogenation of alkenes and an arene, such as benzene. [1]

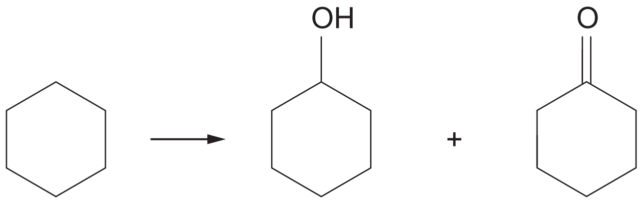
....

1410U401 05

Turn over.

Examiner only

(ii) Cyclohexane, the product from the first stage, is then catalytically oxidised using air. This results in KA (a mixture of cyclohexanol and cyclohexanone). The mixture typically contains around 60% of cyclohexanone.



I. Using characteristic infrared values, describe how you could identify both cyclohexanol and cyclohexanone in KA. You should identify the bonds and their absorption frequencies in your answer. [2]

....

....

....

II. A student suggested that you could find the proportion of cyclohexanol and cyclohexanone in KA by comparing the percentage of oxygen in each compound.

Calculate the percentage of oxygen in each compound. Use your answers to explain why this method would probably not give an accurate answer for the proportion of each compound present in KA. [3]

....

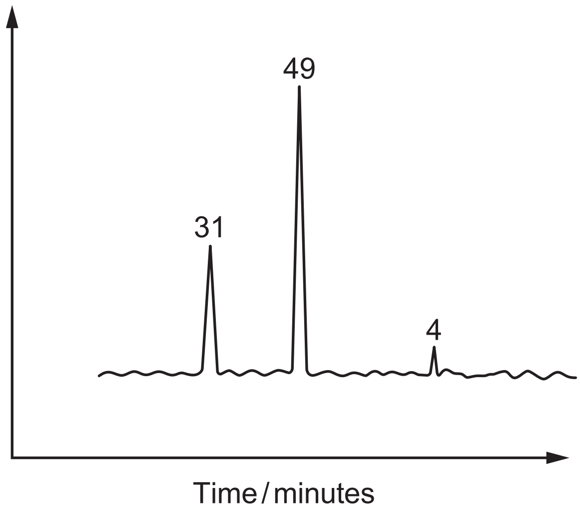
....

....

Examiner only

III. The proportions present in KA can be found more effectively by using gas chromatography.

A typical chromatogram of a sample of KA is shown below. The largest peak is due to cyclohexanone.



Calculate the percentage by volume of cyclohexanone in the mixture. [1]

Percentage of cyclohexanone = ... %

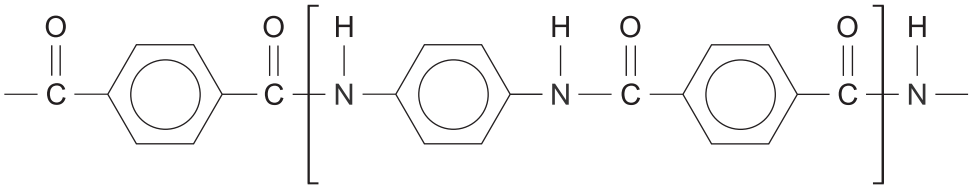
(iii) A sample of the cyclohexanol in KA is oxidised in the laboratory by refluxing with a suitable reagent.

State a suitable oxidising agent that can be used. [1]

....

Examiner only

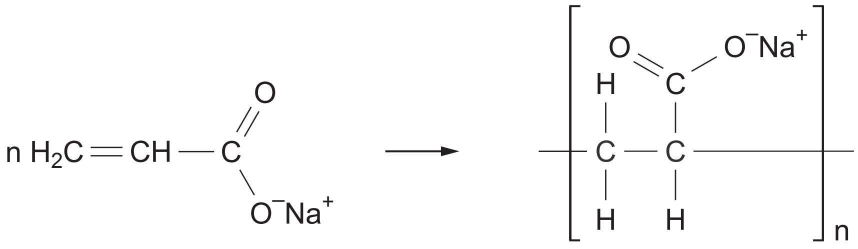
(b) The formula of the polyamide 'Kevlar' is shown below.



Give the structures of two starting materials that can react together to produce this polymer. [2]

Examiner only

(c) The polymer poly(sodiumacrylate) is an example of a super absorbing polymer.



(i) Explain why this polymer is described as an addition polymer and not as a polyester. [2]

....

....

....

(ii) Poly(sodiumacrylate) is capable of absorbing several hundred times its own mass of water, whilst remaining as a solid material.

In an experiment, 4.0×10^{-6} mol of the polymer absorbed 150 g of water. This mass of water is 300 times the mass of the polymer used.

Calculate the relative molecular mass of the polymer. [2]

M_r = ...

14

1410U401 09